

NATIONAL UNIVERSITY OF SINGAPORE

MA3236 - Non-Linear Programming

Midterm Test

(Semester 1 : AY2024/2025)

Date: 02 Oct 2024, Time: 12:15pm - 1:30pm

Student number	
Tutorial group	

INSTRUCTIONS TO CANDIDATES

1. Please write your matriculation/student number only. Do not write your name.
2. This examination paper contains **4** questions and comprises **9** printed pages.
3. Answer **ALL** questions.
4. The total marks for this paper is **50**.
5. This is a CLOSED BOOK examination.
6. Students are allowed to use one A4-sized helpsheet.
7. Candidates may use scientific calculators. Graphical and programmable calculators are not allowed. However, they should lay out systematically the various steps in the calculations.

1. [12 points] Consider the following problem (P):

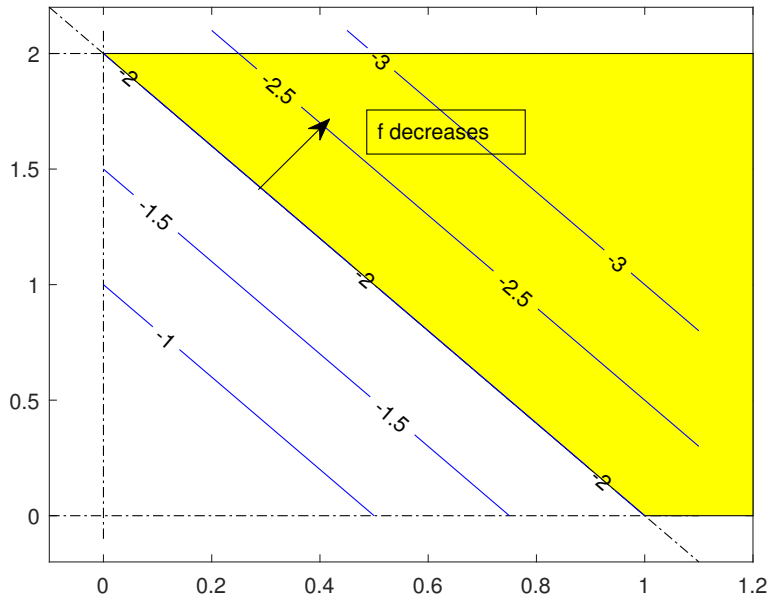
$$(P) \quad \begin{array}{ll} \min_{x,y \in \mathbb{R}} & -2x - y \\ \text{s.t.} & \alpha x + y \leq 2, \\ & 2x + y \geq 2, \\ & x \geq 0, y \geq 0. \end{array}$$

- (a) Determine the value(s) of $\alpha \in [0, 3]$ for which the problem (P) has
- (i) no solution
 - (ii) exactly one solution
 - (iii) infinitely many solution
- (b) For (ii) and (iii) in part (a), give the optimal solution(s) and optimal objective function value(s) in terms of α .

Justify your solution in each case. You may use a graph to illustrate your answer.

Solution.

- (a)(i) (P) has no solution when $\alpha = 0$. Since it gives an unbounded objective function value.



- (a)(ii) (P) has exactly one solution when $\alpha \in (0, 3] \setminus \{2\}$. When $\alpha \in (0, 2)$, the feasible region form a closed triangle. When $\alpha \in (2, 3)$, there is only one feasible point $(0, 2)$.

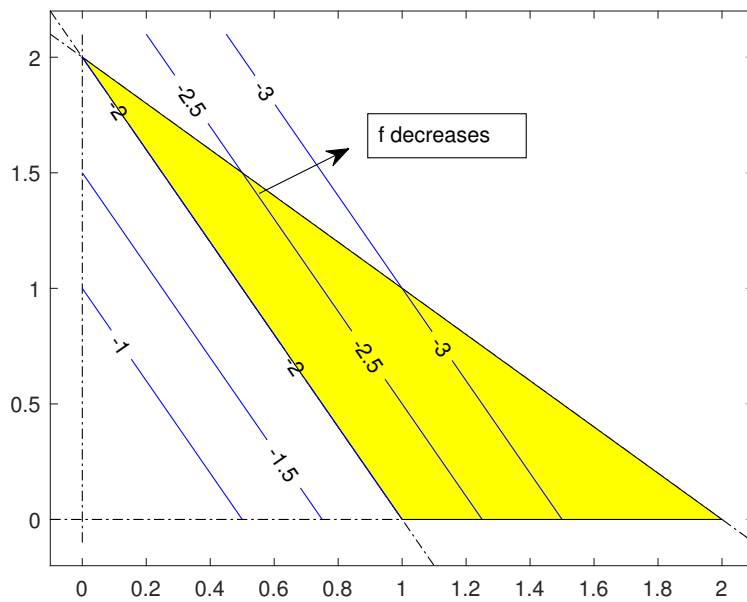


Figure 1: When $\alpha = 1$, the optimal solution is $(2/\alpha, 0)$

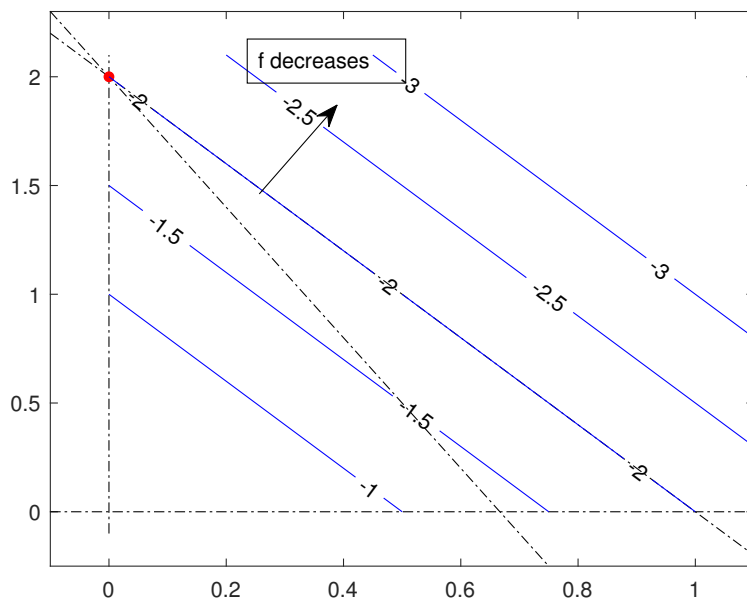
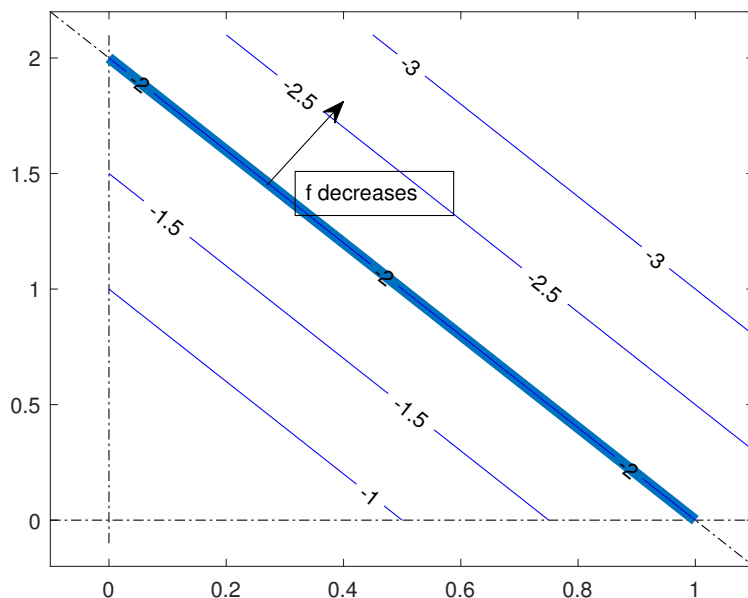


Figure 2: When $\alpha = 3$, the only feasible point is $(0, 2)$.

(a)(iii) (P) has infinitely many solution when $\alpha = 2$, since the constraints give

the line segment between $(0,2)$ and $(1,0)$.



(b) In part (ii), the optimal solution is

$$\begin{cases} \left(\frac{2}{\alpha}, 0\right), & \text{if } \alpha \in (0, 2), \\ (0, 2), & \text{if } \alpha \in (2, 3] \end{cases}$$

The corresponding optimal objective function value is

$$\begin{cases} -\frac{4}{\alpha}, & \text{if } \alpha \in (0, 2), \\ -2, & \text{if } \alpha \in (2, 3] \end{cases}$$

In part (iii), the set of optimal solution is $\{(x, y) \in \mathbb{R}^2 \mid 2x + y = 2, x \geq 0, y \geq 0\} = [(0, 2), (1, 0)]$. The optimal objective function value is -2.

2. [14 points] Consider $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by

$$f(\mathbf{x}) = 3x_1^2 + x_2^2 + x_3^2 - x_2^2 x_3^2$$

- (a) Find all the stationary points of f in $\mathbb{R}^3$.
- (b) For each stationary point obtained in (a), determine whether it is a local maximizer or a local minimizer or a saddle point of f in $\mathbb{R}^3$.
- (c) Does f have a global minimizer in $\mathbb{R}^3$? Justify your answer.

Solution.

(a) $\nabla f(\mathbf{x}) = \begin{pmatrix} 6x_1 \\ 2x_2 - 2x_2 x_3^2 \\ 2x_3 - 2x_2^2 x_3 \end{pmatrix}$. Solving $\nabla f(\mathbf{x}) = 0$ gives us five stationary points: $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix}$

(b) $H_f(0, 0, 0) = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ is a positive definite matrix. This point is a (strict) local minimizer.

$H_f(0, 1, 1) = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 0 & -4 \\ 0 & -4 & 0 \end{pmatrix}$ is an indefinite matrix. This point is a saddle point.

$H_f(0, -1, 1) = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}$ is an indefinite matrix. This point is a saddle point.

$H_f(0, 1, -1) = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}$ is an indefinite matrix. This point is a saddle point.

$H_f(0, -1, -1) = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 0 & -4 \\ 0 & -4 & 0 \end{pmatrix}$ is an indefinite matrix. This point is a saddle point.

Remarks. The eigenvalues of $\begin{pmatrix} 6 & 0 & 0 \\ 0 & 0 & -4 \\ 0 & -4 & 0 \end{pmatrix}$ are 6 and the eigenvalues

of the submatrix $\begin{pmatrix} 0 & -4 \\ -4 & 0 \end{pmatrix}$. For this submatrix, the determinant is negative, implies that there is a positive and a negative eigenvalue. So, we can conclude that the matrix is indefinite.

- (c) There is no global minimizer. Take $x_1 = 0$ and $x_2 = 2$, then

$$f(\mathbf{x}) = 4 - 3x_3^2$$

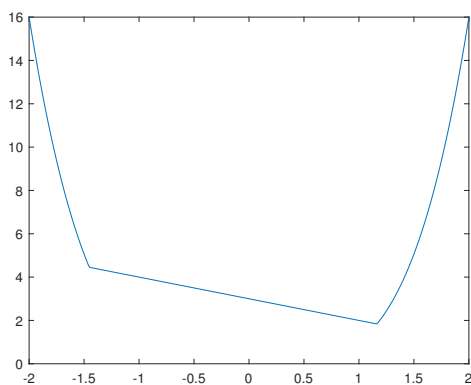
When $x_3 \rightarrow \infty$, $f(\mathbf{x}) \rightarrow -\infty$.

3. **[12 points]** Consider $\min_{x \in \mathbb{R}} f(x) := \max\{x^4, |x - 3|\}$

- Show that f is a unimodal function. You may use a graph to help you illustrate your answer.
- Manually run one iteration of Golden section search to obtain $[a_1, b_1]$, starting with the initial interval $[a_0, b_0] = [0, 3]$.
- Although f is not continuously differentiable, we can try to use Bisection method to get an approximate solution in this case if we choose an appropriate initial interval. Manually run two iterations of Bisection method to obtain $[a_2, b_2]$, starting with the initial interval $[a_0, b_0] = [0, 3]$.
- Can we use Newton method to solve this problem, starting at the initial guess $x_0 = 0$?

Solution.

- By plotting out the graph, we can see that the function has a unique minimizer. On the left of the minimizer, the function is strictly decreasing; on the right of the minimizer, the function is strictly increasing.



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$$\begin{aligned}\lambda_0 &= b_0 - \alpha(b_0 - a_0) = 3 - 3\alpha = 1.1459 \\ \mu_0 &= a_0 + \alpha(b_0 - a_0) = 3\alpha = 1.8541 \\ f(\lambda_0) &= \max\{1.1459^4, |1.1459 - 3|\} = 1.8541 \\ f(\mu_0) &= \max\{1.8541^4, |1.8541 - 3|\} = 11.8177\end{aligned}$$

Since $f(\lambda_0) < f(\mu_0)$, we will take $[a_1, b_1] = [a_0, \mu_0] = [0, 1.8541]$.

- (c) [Iteration 1.] $f'(0) = -1 < 0$ and $f'(3) = 4x^3 = 27.4 > 0$. Since the sign of f' is different at two end points, we can proceed with the Bisection method. Taking the midpoint 1.5, we have $f'(1.5) = 13.5 > 0$. So $[a_1, b_1] = [0, 1.5]$.

[Iteration 2.] Taking the midpoint 0.75, we have $f'(0.75) = -1 < 0$. So $[a_2, b_2] = [0.75, 1.5]$.

- (d) No. $f''(0) = 0$, so we cannot proceed with the Newton iteration.

4. **[12 points]** Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex and continuous function, and that $f(\mathbf{x}) - \|\mathbf{x}\|^2$ is also convex on $\mathbb{R}^n$. Assume that $\mathbf{x}^*$ is a local minimizer of f .

- (a) Show that $\mathbf{x}^*$ is a global minimizer of f .
 (b) Show that $\mathbf{x}^*$ is the unique global minimizer of f .

Solution. Note: f might not be differentiable, so you should not assume ∇f and H_f exist.

- (a) Since f is convex, the local minimizer $\mathbf{x}^*$ is also a global minimizer of f .

- (b) **Solution 1.**

We shall prove by contradiction. Suppose $\mathbf{x}^*$ is not the unique global minimizer, then there exists $\mathbf{y}^* \neq \mathbf{x}^*$ where $f(\mathbf{x}^*) = f(\mathbf{y}^*)$.

Since $f(\mathbf{x}) - \|\mathbf{x}\|^2$ is also convex, considering the midpoint of $\mathbf{x}^*$ and $\mathbf{y}^*$, then we can make use of the definition of convexity to obtain

$$\begin{aligned} f\left(\frac{1}{2}(\mathbf{x}^* + \mathbf{y}^*)\right) - \left\|\frac{1}{2}(\mathbf{x}^* + \mathbf{y}^*)\right\|^2 &\leq \frac{1}{2}(f(\mathbf{x}^*) - \|\mathbf{x}^*\|^2) + \frac{1}{2}(f(\mathbf{y}^*) - \|\mathbf{y}^*\|^2) \\ &= f(\mathbf{x}^*) - \frac{1}{2}\|\mathbf{x}^*\|^2 - \frac{1}{2}\|\mathbf{y}^*\|^2 \end{aligned}$$

This implies

$$\begin{aligned} f\left(\frac{1}{2}(\mathbf{x}^* + \mathbf{y}^*)\right) &\leq f(\mathbf{x}^*) - \frac{1}{2}\|\mathbf{x}^*\|^2 - \frac{1}{2}\|\mathbf{y}^*\|^2 + \left\|\frac{1}{2}(\mathbf{x}^* + \mathbf{y}^*)\right\|^2 \\ &\leq f(\mathbf{x}^*) - \frac{1}{2}\|\mathbf{x}^*\|^2 - \frac{1}{2}\|\mathbf{y}^*\|^2 + \frac{1}{4}\|\mathbf{x}^*\|^2 + \frac{1}{4}\|\mathbf{y}^*\|^2 + \frac{1}{2}(\mathbf{x}^*)^T \mathbf{y}^* \\ &\leq f(\mathbf{x}^*) - \frac{1}{2}\|\mathbf{x}^* - \mathbf{y}^*\|^2 < f(\mathbf{x}^*) \end{aligned}$$

This contradicts with the fact that $\mathbf{x}^*$ is a global minimizer of f . Hence $\mathbf{x}^*$ must be the unique global minimizer of f .

Solution 2.

Since $\|\mathbf{x}\|^2$ is a strictly convex function, we have $(f(\mathbf{x}) - \|\mathbf{x}\|^2) + \|\mathbf{x}\|^2 = f(\mathbf{x})$ is also a strictly convex function. Hence, a global minimizer of a strictly convex function must be unique.

End of paper